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## Digital Sovereignty in the Era of Industry 5.0: Challenges and Opportunities

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### Abstract

This paper examines the core concept of digital sovereignty within the Industry 5.0 framework, highlighting the importance of sovereign control over data, digital infrastructure, and digital artifacts such as algorithms, models, and visualizations. Trustworthy, autonomous, distributed architectures are essential to building resilience and independence in a multipolar world. Through a comprehensive analysis of existing literature, this paper identifies key challenges and opportunities in achieving digital sovereignty, using a PRISMA chart to select relevant studies from six databases systematically. The article offers strategic insights on extending digital technologies to ensure digital sovereignty while encouraging independent and ethical innovation. By identifying research gaps and proposing future study areas, this work aims to inspire further exploration and development in securing digital sovereignty, supporting resilient digital value chains, and advancing the goals of Industry 5.0.

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### 1. Introduction

Digital sovereignty denotes the capability of a nation, institution, or personnel to control and supervise its digital infrastructure, data, and technology free from excessive reliance on external commodities or foreign forces. It assures the governance of digital resources, containing data storage, internet infrastructure, software, and digital services, to safeguard national security, confidentiality, economic interests, and cultural values [1]. Proactively safeguarding their inhabitants' and economy's interests in an increasingly digitally globalized world, several nations are placing a greater

emphasis on digital sovereignty to preserve control over vital digital assets. The key aspects of digital sovereignty include a) Data sovereignty, b) Technological autonomy, c) Cybersecurity, and d) Regulatory control. Data sovereignty assures that data produced within a nation's territory is kept, processed, and controlled by the local ruling, obstructing foreign access or authority over such data. Technological autonomy diminishes dependence on foreign technological suppliers or media, which may present hazards if other nations' legislation, guidelines, or sanctions govern those suppliers. Cybersecurity safeguards nationwide electronic infrastructure against outside interference, including cyberattacks, spying, and manipulation. Protecting individuals' privacy, ensuring intellectual property ownership, and policing inappropriate content online are all rules and regulations that fall under the purview of regulatory authority. The advent of digitalization and subsequent technological advances, particularly in artificial intelligence (AI), has become a powerful force that goes beyond economics and geopolitics [2]. The realization of data's power has coincided with the rapid pace of digitalization or digital transformation, as an intricate web of entities gathers, processes, and exchanges data to unlock its worth [3]. Due to this change, there are concerns about privacy and sovereignty since, despite the large amounts of personal data collected, individuals typically do not own or have control over them. Data governance, cybersecurity management, and privacy protection are essential for controlling who has access to how this data is used and for protecting one's digital assets and identity [4].

Industry 5.0 highlights the integration of human imagination, values, and talents alongside cutting-edge technologies like automation, artificial intelligence (AI), and the Internet of Things (IoT), unlike Industry 4.0, which concentrates heavily on automation, smart factories, and the seamless connection between machines [5], Industry 5.0 highlights human-machine collaboration and aims to create more sustainable, personalized, and human-centric manufacturing systems [6]. Fig. 1 illustrates the progression of the industrial process. The initial industrial process was marked by mechanization, whereas the subsequent introduced mass manufacturing, automation, and electrification [7]. Industry 3.0 integrated information technology and automation [8]. Industry 4.0 incorporates sophisticated technology into industrial processes, including the Internet of Things (IoT), artificial intelligence (AI), robotics, and big data analytics. It encompasses the real-time oversight and administration of machinery and equipment throughout all aspects of the industrial process [9].

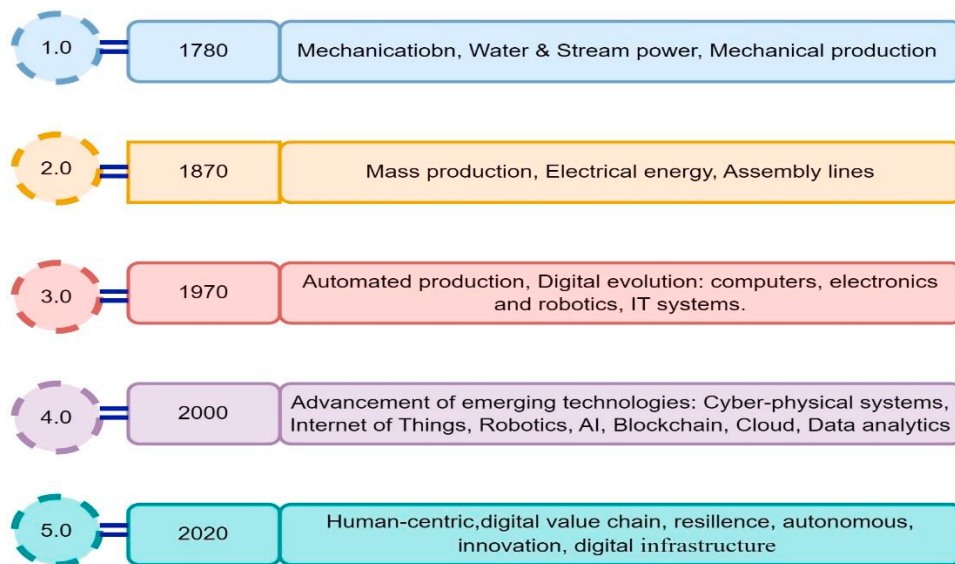


Fig. 1. Development of industrial evolution.

Industry 5.0 is characterized by human-centric, resilient, and sustainable systems, with digital sovereignty as a key factor in enabling secure and autonomous innovation. Digital sovereignty is becoming increasingly important for nations, industries, and individuals in the context of rapidly advancing autonomous technologies. The review addresses the primary aspects, opportunities, and potential strategic frameworks for resilient and self-sovereign supply chains, industrial ecosystems, and manufacturing processes in Industry 5.0. The study aims to address the following

research questions.

RQ1. What is digital sovereignty in an Industry 5.0 context?

RQ2. How can digital sovereignty be preserved while ensuring resilient and self-sovereign supply chains, industrial ecosystems, and manufacturing processes?

RQ3. What are the opportunities created by advancing digital technologies to foster digital sovereignty?

Addressing the identified research questions furnishes information to promote comprehending the digital sovereignty of the industry 5.0 research domain, facilitate cross-domain study techniques, and supply visions for potential study directions.

The remaining study is organized as follows: Section 2 illustrates the systematic literature review of methodology. Section 3 depicts the outcomes analysis. Section 4 presents the study's discussion, limitations, open gaps, and future research directions. Finally, Section 5 concludes the paper.

## 2. Methodology

Literature reviews serve as effective sources for knowledge creation. They systematically gather existing scholarly work and employ straightforward explicit and implicit data synthesizing to address identified research questions. A systematic literature review (SLR) is a comprehensive, evident, and replicable approach to selecting and analyzing scientific publications, testifying search to determine previous research. Table 1 illustrates the SLR protocol employed in this work, and Fig. 2 exhibits the PRISMA methods used for selecting identified works.

The PRISMA framework, a comprehensive process, begins with an initial collection of articles and a starting point collection found using keyword searches in specialized scientific publication databases [10]. The sample is meticulously finalized using inclusion and exclusion elements, forming the intermediate instance. Backward searches are then employed to incorporate articles not discovered in the initial searches, ensuring the final sample designated for analysis, referred to as the synthesis sample, is of the highest quality. The synthesis sample is further examined via descriptive and thematic analyses, providing potent answers to the research questions. In October 2024, after applying search rules to particular publication records, 299 articles were found in the initial sample. 121 articles were found in the synthesis record following the exclusion rules, and 33 were discovered using inclusion rules and reference assessment.

Table 1. SLR protocol.

| Elements           | Details                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|
| SLR                | [11]                                                                                                                     |
| Analytical sample  | Published exiting studies                                                                                                |
| Data sources       | ScienceDirect, IEEE Xplore, Springer, Scopus, Google Scholar. Taylor and Francis.                                        |
| Search combination | (digital sovereignty OR data sovereignty OR technological sovereignty) AND (Industry 5.0 OR fifth industrial revolution) |
| Exclusive rule     | a)Scientific article b)Study relevant c) in English language                                                             |
| Inclusive rule     | Title or abstract or related to digital sovereignty and Industry 5.0                                                     |



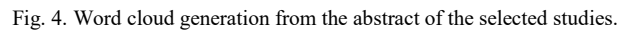


Table 2. Key aspects of sovereignty in Industry 5.0.

| Sr. No | Areas                           | Description                                                                                                                                                                                                                   | Samper papers |
|--------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1      | Data Sovereignty                | It identifies the owner of and governs data within a highly interconnected environment. Abundant data from AI, IoT, and robotics raises issues regarding safeguarding national and industrial information in Industry 5.0.    | [12]          |
| 2      | Cross-border data movement      | Facilitating data movement between many countries while adhering to national and international laws poses considerable difficulty.                                                                                            | [13]          |
| 3      | Data privacy                    | Protecting information containing personal, private, or functional data from unwanted access or exploitation is challenging.                                                                                                  | [14]          |
| 4      | Increasing threat landscape     | Incorporating emerging technologies like AI, IoT, and cyber-physical systems makes the sector increasingly susceptible to cyberattacks, which may devastate national or industrial sovereignty.                               | [15]          |
| 5      | Insufficient Security Mechanism | Several businesses may need enhanced cybersecurity measures to avoid, identify, and respond to threats in real time.                                                                                                          | [16]          |
| 6      | Fragmented standards            | The absence of uniform standards among nations and sectors can impede system integration, diminish efficiency, and complicate international industrial operations.                                                            | [17]          |
| 7      | Technology fragmentation        | Developing proprietary technologies or standards in various regions often exacerbates this, resulting in a fragmented digital landscape that hinders collaboration and technological integration.                             | [18]          |
| 8      | Insufficient Regulations        | Different data protection regulations, AI ethics frameworks, and industry norms must be addressed.                                                                                                                            | [19]          |
| 9      | Cross-border law                | Conflicts over jurisdiction and applying laws about data, technology, and intellectual property may emerge as industrial systems become increasingly globally integrated, potentially impacting the industry.                 | [20]          |
| 10     | Authenticity and Ethics in AI   | The growing application of AI in decision-making processes has raised concerns about algorithmic discrimination, righteousness, and transparency. AI systems should be designed to respect moral principles and human values. | [21], [22]    |
| 11     | Human-centric approach          | Industry 5.0, with its strong emphasis on human-machine collaboration, is a beacon of hope. However, navigating the social implications of automation's effects on employment and economic inequality may be challenging.     | [23], [24]    |
| 12     | Public Confidence               | Developing and preserving trust is essential for the general adoption and acceptance                                                                                                                                          | [19], [25]    |



|    |                             |                                                                                                                                                                                              |            |
|----|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|    |                             | of digital systems, especially when those systems make important decisions.                                                                                                                  |            |
| 13 | Technological Resilience    | Sustaining sovereignty requires ensuring that vital infrastructure—like cloud platforms or AI-driven manufacturing systems—withstands disruptions, cyberattacks, and technological failures. | [26], [27] |
| 14 | Responsibility and Openness | The ongoing challenge of regulating AI systems in Industry 5.0, especially regarding accountability for AI-generated decisions, is a complex and evolving issue.                             | [28]       |
| 15 | Decision-making             | AI systems are taking greater autonomous roles, and human oversight and control over important decision-making processes are important to preserving sovereignty in industrial operations.   | [29]       |

The key aspects of digital sovereignty in Industry 5.0 are presented graphically, as pictured in Fig. 5.

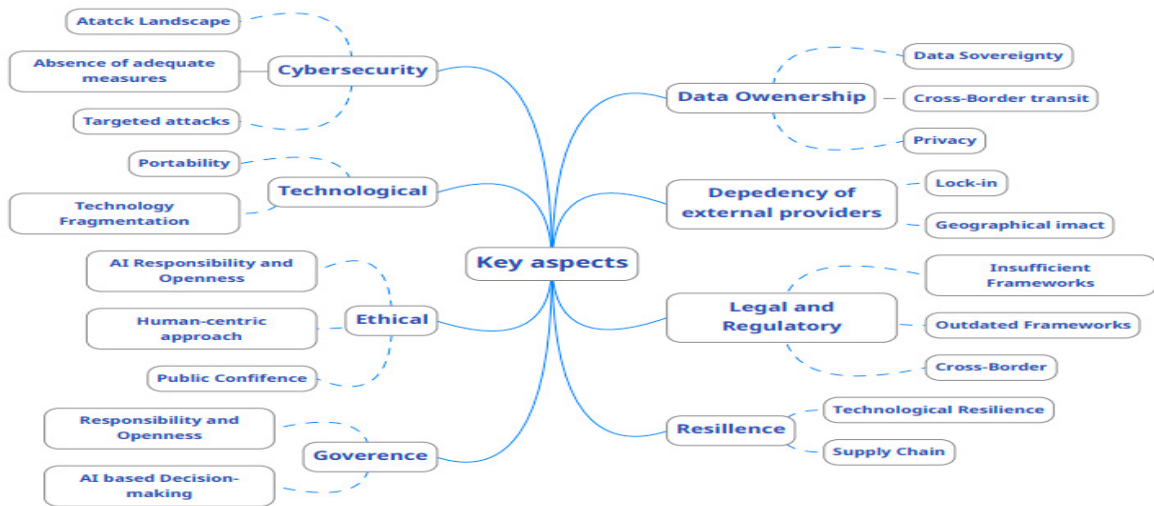


Fig. 5. Key aspects of digital sovereignty in an Industry 5.0 context.

The second research question is: “How can digital sovereignty be preserved while ensuring autonomous and ethical innovation?” Maintaining digital sovereignty while fostering autonomous and ethical innovation necessitates a comprehensive strategy reconciling technological independence, ethical principles, and international collaboration. Table 3 demonstrates the key strategies to achieve this. Fig. 6 illustrates the outlines of strategies to preserve digital Sovereignty.

Table 3. Strategies to preserve digital sovereignty.

| Sr. No | Areas                           | Descriptions                                                                                                                                                                                                                                                                                        | Samper papers |
|--------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1      | Data Sovereignty and Protection | Establish robust data sovereignty legislation and promote advancing technologies emphasizing user privacy, transparency, and data protection.                                                                                                                                                       | [30]          |
| 2      | Develop infrastructure          | Countries should invest in developing and maintaining digital infrastructure, including national data centers and communication networks. Local governance of infrastructure facilitates autonomous technological development and promotes innovation.                                              | [31]          |
| 3      | Open standard and technology    | Support for open-source software and open standards enhances technological autonomy and diminishes reliance on proprietary systems governed by foreign entities. Open-source initiatives promote openness and accountability, enabling more inclusive innovation that adheres to ethical standards. | [32]          |
| 4      | Ethics and Regulation           | Develop national legislation to reflect local values, defend public interests, and prevent foreign firms from using AI and new technology. Encourage honest, fair, ethical AI and digital technology development. Include frameworks for algorithmic accountability and fairness.                   | [33]          |

|   |                                |                                                                                                                                                                                                                                             |      |
|---|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5 | Cybersecurity and Resilience   | Enhance national cybersecurity capabilities to safeguard key infrastructure, data, and digital assets from external threats. Allocate resources to cybersecurity research and innovation, promoting novel tools and technology advancement. | [34] |
| 6 | Literacy and Education         | Invest in digital literacy initiatives that enable citizens to comprehend the significance of digital sovereignty, privacy, and data protection.                                                                                            | [35] |
| 7 | Sustainability and development | Assure that national technology strategies are congruent with sustainable development objectives. Promote innovation that is both technologically sophisticated and beneficial to societal welfare.                                         | [36] |
| 8 | Partnership                    | Facilitate public-private collaborations that correspond with national interests, allowing government control while encouraging innovation in the private sector.                                                                           | [37] |

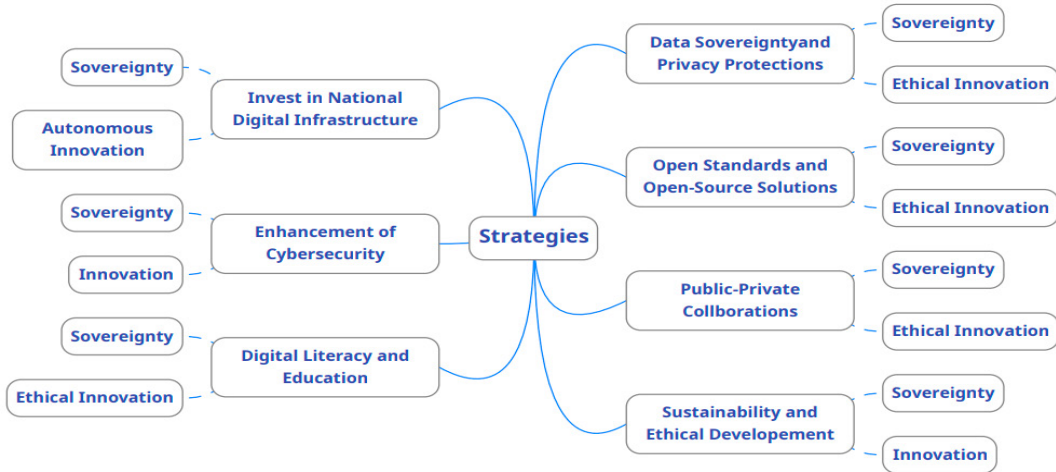


Fig. 6. Overview of strategies to preserve digital sovereignty.

The third research question is: “What are the opportunities created by advancing digital technologies to foster digital sovereignty?” The advancement of digital technologies generates multiple opportunities to promote digital sovereignty. This enables nations and entities to retain authority over digital assets while fostering innovation. These technologies safeguard national interests, stimulate local economic development, and diminish dependence on external systems. Table 4 demonstrates the opportunities to advance Digital Sovereignty, illustrated graphically in Fig. 7.

Table 4. Opportunities to advance digital sovereignty.

| Sr. No | Areas                                | Opportunities                                                                   | Reference |
|--------|--------------------------------------|---------------------------------------------------------------------------------|-----------|
| 1      | Data Sovereignty                     | Cloud computing, data analytics, blockchain                                     | [38]      |
| 2      | Autonomous Infrastructure            | Cloud computing, edge computing, and 5G networks.                               | [39]      |
| 3      | Cybersecurity Innovations            | AI-driven threat detection and secure communication using blockchain.           | [40]      |
| 4      | Smart Governance                     | e-governance and digital public services                                        | [41]      |
| 5      | Self-sovereign infrastructure        | biometric verification or blockchain-based ID systems                           | [42]      |
| 6      | AI and Automation                    | Optimize enterprises in areas such as farming, manufacturing, and logistics.    | [43]      |
| 7      | Distributed sovereign infrastructure | Cloud infrastructure is geographically distributed.                             | [13]      |
| 8      | Trustworthy                          | Respect for human autonomy, prevention of threats, fairness, and explicability. | [44]      |



**Fig. 7.** Opportunities to foster digital sovereignty.

#### 4. Discussion

The study addresses research question 1 (RQ1) by systematically demonstrating the key aspects of digital sovereignty in Industry 5.0, as demonstrated in Table 2 and Fig. 5. The second research question (RQ2) is addressed by highlighting extensive strategies to preserve digital sovereignty and ensure autonomous and ethical innovation, as depicted in Table 3 and Fig. 6. The third research question (RQ3) is addressed by highlighting the opportunities created by advancing digital sovereignty, as depicted in Table 4 and Fig. 7.

Digital sovereignty in Industry 5.0, especially with the pervasive application of artificial intelligence (AI), pertains to empowering companies and nations to control and protect their digital assets, data, and technology. Implementing digital sovereignty in Industry 5.0 necessitates handling critical data privacy, autonomy, resilience, and moral AI issues. Industry 5.0 emphasizes human-centric innovation, incorporating AI in manners that prioritize human welfare, sustainability, and collaborative technology. This allows nations and commodities to implement AI judiciously while maintaining sovereignty and oversight over digital resources and technologies.

The timing of the study may affect the study's validity due to the swift advancement of digital sovereignty and Industry 5.0. This indicates that the study may only partly contain recent references and improvements. Moreover, the survey design, encompassing the process for data aggregation and the criteria for including or excluding retrieved articles, may influence the study's internal validity. Sampling bias may arise when the data sample fails to represent the complete population within the research domain accurately.

Future studies of digital sovereignty within Industry 5.0 should consider the intricate interactions among technology, ethics, security, sustainability, and international collaboration. By exploring these avenues, nations can enhance their comprehension of maintaining digital governance while fostering innovation, upholding ethical standards, and driving through the changing geopolitical environment.

#### 5. Conclusion

Digital sovereignty recreates a significant role in the future of Industry 5.0, portrayed by sovereign control over digital technologies and data supporting global industrial ecosystems, value chains, and manufacturing processes. In a multipolar world, it's required to de-risk global supply chains and mission-critical digital infrastructures. There is a need to increase trust in technology, data, and business entities that support global logistics operations and manufacturing processes. To achieve resilience in these operations and processes, the level of autonomy plays a key



role reflected in distributed and autonomous digital architectures and ecosystems. The article additionally exemplifies the in-depth study of the possibilities of expanding digital technologies to facilitate digital sovereignty.

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